Noise, one step at a time

An image model learns by watching pictures dissolve into static, one small step at a time, and a shortcut jumps to any step at once.

Based on Denoising Diffusion Probabilistic Models, Ho et al., 2020

1 Start with data

A diffusion model learns to make data, such as pictures, by first learning how it gets destroyed. The destroying half, the *forward process*, has nothing to learn: it is a fixed recipe that adds noise, a little at a time. Ho, Jain and Abbeel used it for images in denoising diffusion probabilistic models, DDPM for short. Every pixel follows the same rule, so one number is enough to watch it work.

Our data is 2,000 numbers drawn from two clumps, one around −2 and one around +1.5, each with a spread of 0.25. Think of them as two kinds of picture. The question is what the noise does to the clumps, step by step.

2 One small step

Each step shrinks the current value a little, then adds a little fresh random noise, ε, drawn from a bell curve centred on 0 with a spread of 1. One small number for each step, β_t, sets how much of each. The shrink matters. It makes exactly enough room for the new noise, so values whose spread is already 1 keep a spread of 1. Without the shrink, noise would pile up and the values would drift ever wider.

$$x_t = \sqrt{(1 - \beta_t)} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \varepsilon$$

If x_{t-1} has variance 1, the new value has variance $(1 - \beta_t) + \beta_t = 1$: the shrink makes room for exactly the noise that is added.

The step sizes β_t follow a schedule. DDPM uses $T = 1{,}000$ steps, with β rising in equal increments from 0.0001 at the first step to 0.02 at the last. Early steps barely touch the data; the last steps are 200 times larger.

3 Jump to any step

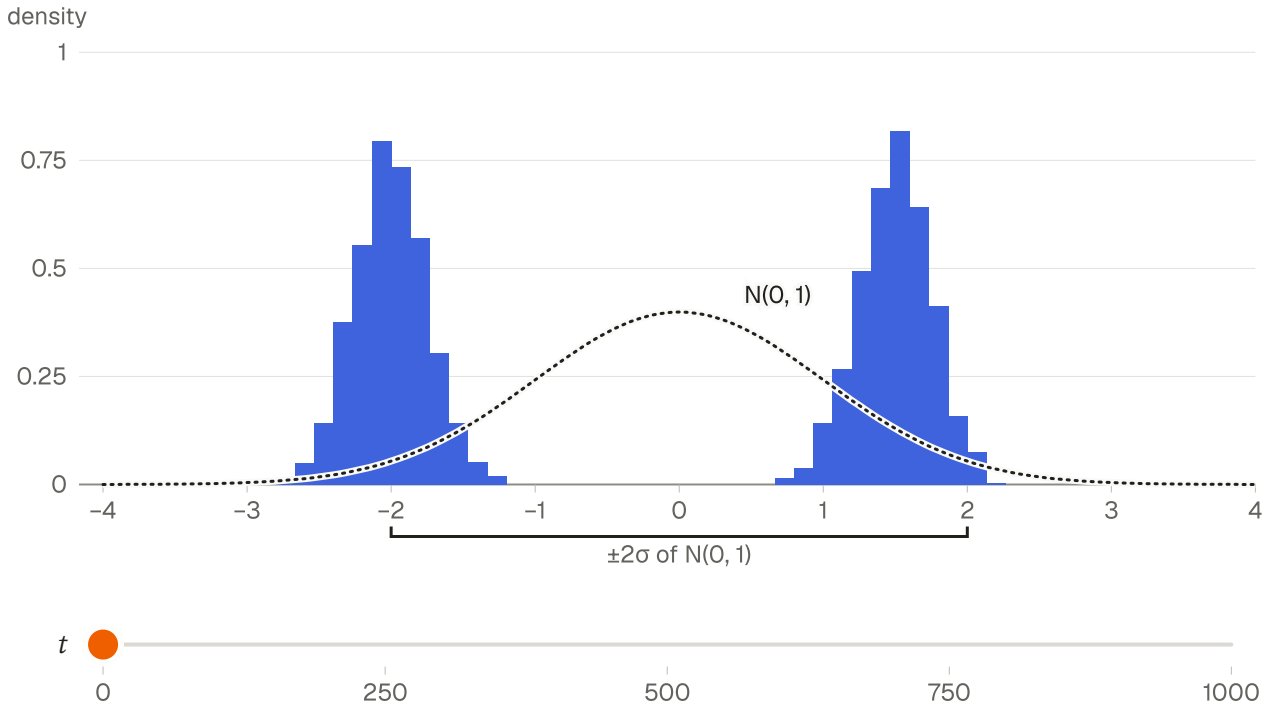
Steps add up, and there is a shortcut. Multiply the numbers $1 - \beta_t$ of every step so far, call the product $\bar{\alpha}_t$, and the whole chain from the starting value x_0 collapses into one line: keep part of the original value, then add one draw of noise. The first part, $\sqrt{\bar{\alpha}_t}$, is how much of the original signal is left; the second is how much noise has replaced it.

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{(1 - \bar{\alpha}_t)} \cdot \varepsilon, \text{ with } \bar{\alpha}_t = (1 - \beta_1)(1 - \beta_2)\ldots(1 - \beta_t)$$

One draw of noise ε stands in for all the small ones, because the sum of independent bell-curve noises is again a bell curve.

The figure uses that line directly. Drag t from 0 to 1,000. At $t = 100$ the clumps are softer but still apart. Around $t = 370$ they merge into one hump. By $t = 600$ only 16% of the signal is left and the samples sit on the dotted bell curve of pure noise; at $t = 1{,}000$ their spread is 1.01 and the data is gone.

Two clumps become one bell



signal $\sqrt{\bar{\alpha}_t}$ **1.000** noise $\sqrt{(1-\bar{\alpha}_t)}$ **0.000** spread (SD) **1.77**

Shown at $t = 0$. Interactive version at explorable.geekplux.com/s/diffusion#fig-histogram

2,000 samples (seed 2020) pushed through DDPM's linear schedule, β from 0.0001 to 0.02 over 1,000 steps. Each histogram is the exact distribution at that step; each sample keeps one fixed noise draw, so its path between steps is illustrative.

4 Signal fades, noise rises

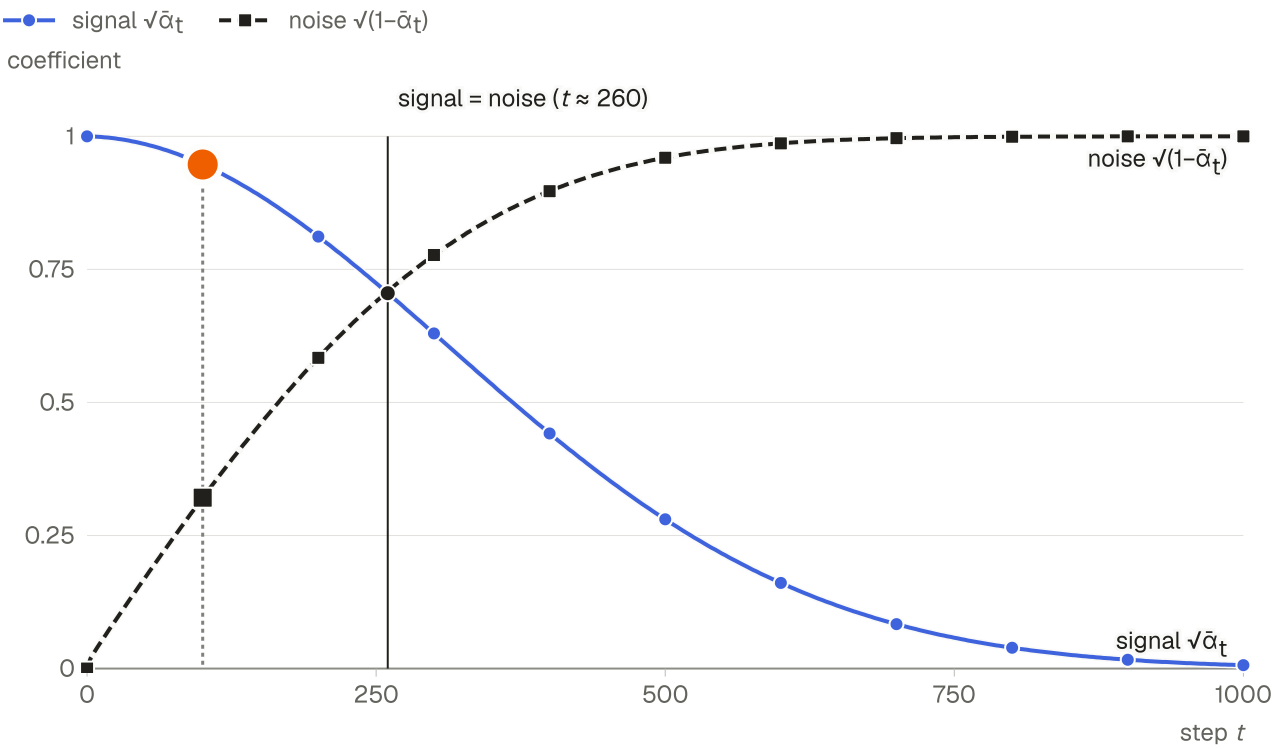
Plot the two parts against t and the schedule shows its shape. The signal falls slowly at first, then quickly, then hardly at all; the noise mirrors it. They cross at $t \approx 260$, where $\bar{\alpha}_t = 0.5$: signal and noise are equally strong, a signal-to-noise ratio of exactly 1, or 0 decibels (dB).

$$\mathrm{SNR} = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$$

The signal's share of the variance divided by the noise's share; in decibels it is 10 times its base-10 logarithm, so a ratio of 1 is 0 dB.

Most of the change happens in the first half. Past $t \approx 750$ the signal is under 6%, and each further step changes little. Later papers proposed other schedules for exactly this reason: the schedule is a design choice, not a law.

How much signal is left at each step



ᾱt **0.897** SNR **9.4** dB

Shown at $t = 100$. Interactive version at explorable.geekplux.com/s/diffusion#fig-signal-noise

Both curves come from the same schedule as the figure above, β from 0.0001 to 0.02. Drag along the signal curve; the readout shows $\bar{\alpha}_t$ and the signal-to-noise ratio in decibels.

5 Why the shortcut matters

Making new data runs the process backwards: a neural network learns to guess the noise ε that was mixed into the value at step t, so it can take a little of it away at each step. Training needs examples at random steps, and the shortcut makes each one cheap: pick a starting value, pick a step t, draw the noise and mix them in one line, with no loop over the earlier steps. DDPM scores the network's guess with one simple number: the squared difference between the true noise and the guessed one.

$$L = \|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t}\cdot x_0 + \sqrt{(1-\bar{\alpha}_t)}\cdot\varepsilon,\ t)\|^2$$

The network ε_θ sees the noisy value and the step t, and is scored by the squared distance between its guess and the noise ε that was really added.

Takeaway

The forward process is a fixed schedule of tiny steps that each shrink the value and add noise. Together they collapse into a one-line jump to any step, and by step 1,000 any data has become pure noise.

Read the original paper